INTERNSHIP PROJECT REPORT

E-Commerce Sales Analysis and Customer Segmentation

Using RFM Analysis and K-Means Clustering

Submitted By

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Internship Organization

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Declaration

I hereby declare that the project entitled **"E-Commerce Sales Analysis and Customer Segmentation"** is my original work completed as part of my internship project.

The analysis, data preprocessing, exploratory data analysis, RFM analysis, customer segmentation, visualizations, dashboard, and business recommendations presented in this report were carried out using Python and relevant data analysis and machine learning techniques.

I further declare that this report has not been submitted elsewhere for the award of any academic or professional qualification.

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Signature: ______________________________

Date: ______________________________

Acknowledgement

I would like to express my sincere gratitude to everyone who supported me during the completion of this internship project.

I am thankful to the internship organization, Thiranex, for providing me with the opportunity to work on a practical data analytics project and develop my technical and analytical skills.

I would also like to thank my mentors, faculty members, and colleagues for their guidance and valuable feedback throughout the project.

Finally, I would like to thank everyone who contributed directly or indirectly to the successful completion of this project.

Abstract

This project focuses on analyzing e-commerce transaction data to understand sales performance, customer purchasing behavior, and customer value.

The project uses an Online Retail transaction dataset containing information about invoices, products, quantities, prices, customers, dates, and countries. The raw data was cleaned and transformed using Python and Pandas. Exploratory Data Analysis (EDA) was then performed to identify sales trends, top-performing products, important markets, and purchasing patterns.

To understand customer behavior, RFM analysis was applied using Recency, Frequency, and Monetary value. Customers were subsequently segmented using the K-Means clustering algorithm. The resulting segments were analyzed according to their purchasing behavior and monetary contribution.

A business dashboard was also developed to present key performance indicators, sales trends, product performance, geographic performance, and customer segments.

The analysis identified high-value customers, inactive customers, important geographic markets, and seasonal sales patterns. Based on these findings, recommendations were developed for customer retention, customer re-engagement, marketing optimization, and revenue growth.

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1. Introduction

1.1 Background

E-commerce businesses generate large volumes of transactional data through customer purchases. This data contains valuable information about products, customers, purchasing frequency, transaction value, and market performance.

However, raw transaction data alone does not provide meaningful business insights. Data analysis techniques are required to transform transaction records into useful information that can support business decision-making.

Customer segmentation is particularly important because different customers have different purchasing behaviors and values. By identifying groups of customers with similar characteristics, businesses can develop more targeted marketing and retention strategies.

This project combines sales analysis with customer segmentation to provide a complete view of e-commerce business performance.

1.2 Problem Statement

The primary problem addressed in this project is the difficulty of extracting actionable business insights from a large volume of e-commerce transaction records.

The project aims to answer questions such as:

- Which products generate the highest sales?
- Which countries contribute the most revenue?
- How does revenue change over time?
- Which customers are the most valuable?
- Which customers are becoming inactive?
- How can customers be grouped according to purchasing behavior?
- What business strategies can be developed from these insights?

1.3 Project Objectives

The main objectives of this project are:

1. To clean and preprocess e-commerce transaction data.
2. To perform exploratory data analysis.
3. To analyze sales trends and product performance.
4. To calculate customer-level RFM metrics.
5. To segment customers using K-Means clustering.
6. To validate and interpret the customer clusters.
7. To develop a business-focused dashboard.
8. To generate actionable business recommendations.

1.4 Project Scope

The project covers the complete data analytics pipeline from raw transaction data to business recommendations.

The scope includes data cleaning, feature engineering, visualization, customer analytics, machine learning-based segmentation, dashboard development, and business interpretation.

2. Dataset Description

2.1 Dataset Overview

The project uses an Online Retail transaction dataset containing e-commerce transactions.

Each record represents a product-level transaction associated with an invoice.

The major fields are shown in Table 2.1.

Table 2.1: Dataset Variables

Column	**Description**
InvoiceNo	Unique invoice or transaction number
StockCode	Unique product identification code
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Date and time of transaction
UnitPrice	Price per unit
CustomerID	Unique customer identification number
Country	Customer's country

2.2 Dataset Characteristics

The dataset contains a large number of transaction records and requires preprocessing before analysis.

Several issues were identified during the initial data inspection, including missing customer identifiers, missing product descriptions, duplicate records, negative quantities, and invalid transaction values.

2.3 Revenue Calculation

Transaction revenue was calculated using:

$$Revenue = Quantity \times UnitPrice \tag{2.1}$$

This derived feature was used throughout the sales analysis.

3. Tools and Technologies

3.1 Python

Python was used as the primary programming language because of its strong ecosystem for data analysis, visualization, and machine learning.

3.2 Pandas

Pandas was used for:

- Data loading
- Data cleaning
- Data transformation
- Grouping and aggregation
- RFM calculation

3.3 NumPy

NumPy was used for numerical transformations and mathematical operations.

3.4 Matplotlib

Matplotlib was used to create sales and customer analytics visualizations.

3.5 Scikit-learn

Scikit-learn was used for:

- Feature scaling

- K-Means clustering
- Silhouette score calculation

3.6 Jupyter Notebook

Jupyter Notebook was used as the primary development environment, allowing data processing, visualization, analysis, and documentation to be performed in an interactive environment.

4. Data Cleaning and Preprocessing

4.1 Initial Data Inspection

The dataset was first inspected to understand its structure, data types, missing values, duplicate records, and potential anomalies.

The following variables were examined:

- Number of rows and columns
- Data types
- Missing values
- Duplicate records
- Numerical statistics

4.2 Missing Values

Missing values were identified primarily in the `Description` and `CustomerID` columns.

Customer-level analysis requires a valid CustomerID. Therefore, records without customer identification were excluded from the customer segmentation analysis.

4.3 Duplicate Records

Duplicate transactions were identified and removed where appropriate to reduce the possibility of duplicated business activity affecting the analysis.

4.4 Invalid Transactions

Transactions with invalid quantities or prices were examined and removed according to the cleaning strategy used in the project.

Cancelled or return transactions represented by negative quantities were treated separately from normal sales transactions.

4.5 Feature Engineering

Several additional features were created:

- Revenue
- Month
- Weekday
- Hour

These features enabled detailed temporal and sales analysis.

5. Exploratory Data Analysis

5.1 Overview

Exploratory Data Analysis was performed to understand sales behavior, product performance, geographic contribution, and temporal patterns.

5.2 Monthly Revenue Analysis

Monthly revenue was calculated by grouping transaction revenue by month.

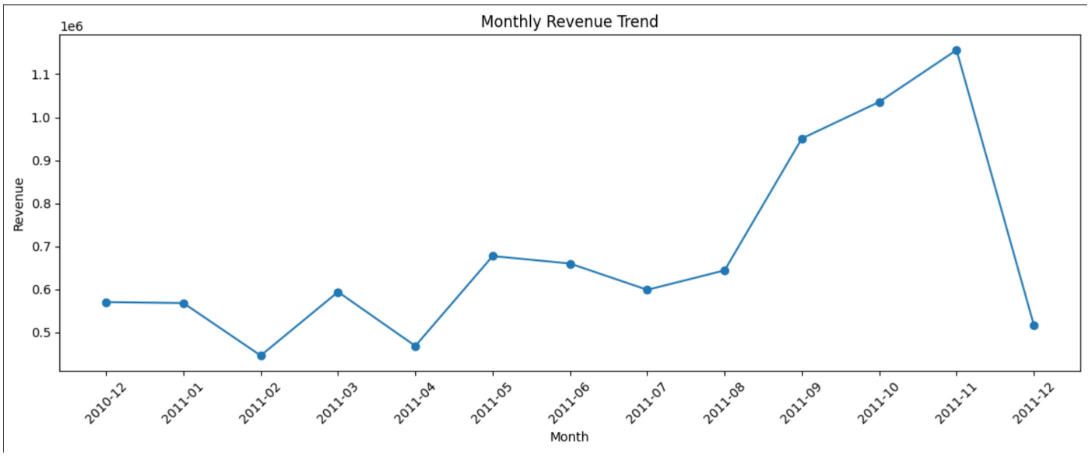

Figure 5.1: Monthly Revenue Trend

The analysis showed significant variation in monthly revenue, with strong growth toward the later months of the analyzed period.

5.3 Top Products

The top products were identified based on total quantity sold.

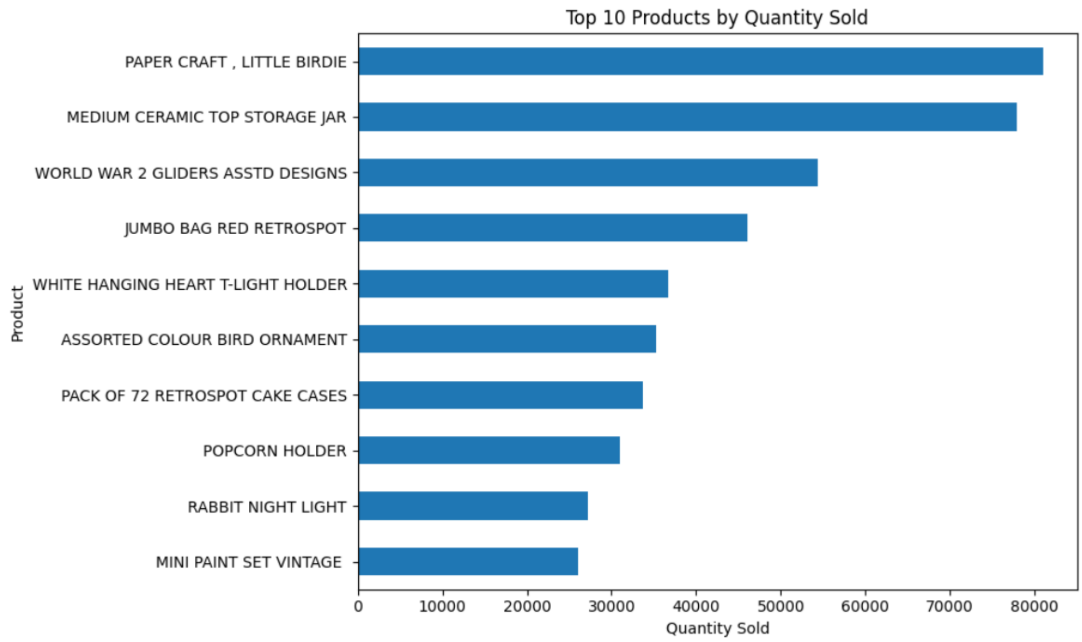

Figure 5.2: Top 10 Products by Quantity Sold

The results show that a relatively small group of products accounts for a substantial amount of sales volume.

5.4 Revenue by Country

Revenue was analyzed across countries.

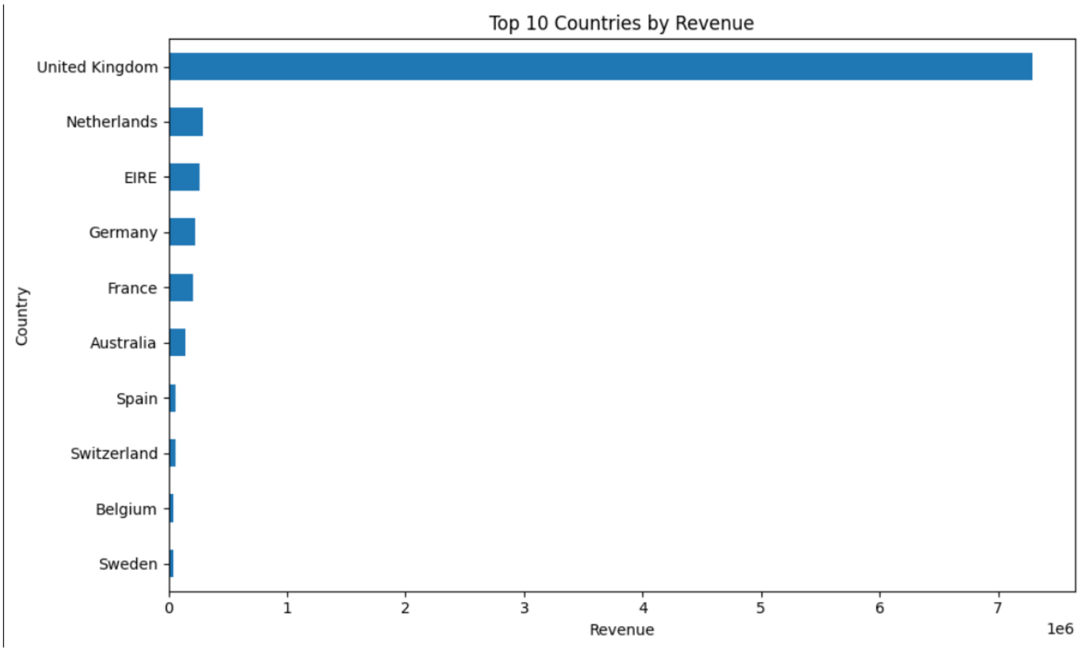

Figure 5.3: Top 10 Countries by Revenue

The United Kingdom was the dominant market in terms of revenue.

This indicates that the business has a strong concentration of sales in the UK market.

5.5 Revenue by Weekday

Revenue was also analyzed by day of the week.

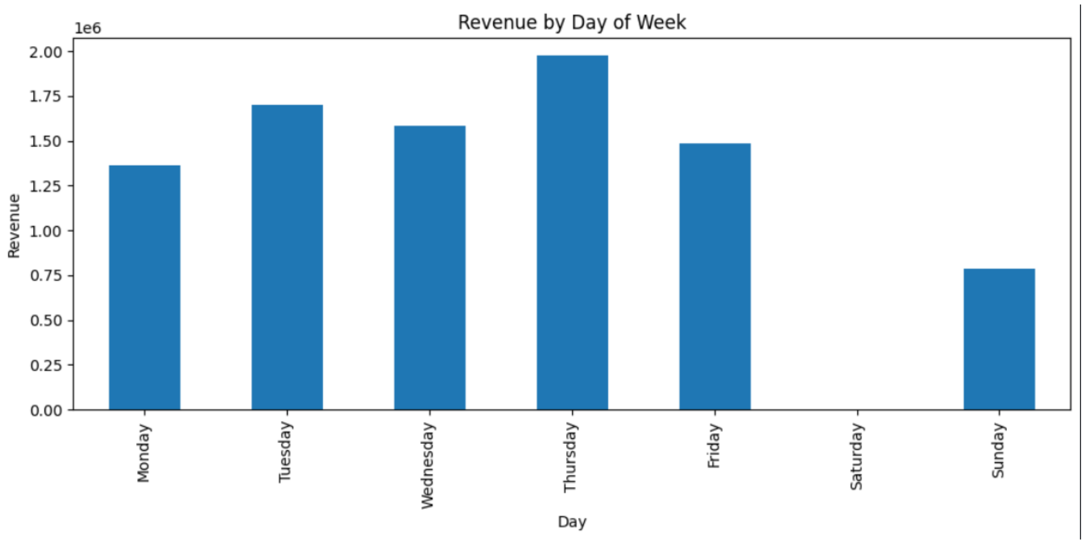

Figure 5.4: Revenue by Day of Week

Thursday was identified as the strongest revenue-generating weekday in the analyzed data.

5.6 Revenue by Hour

Hourly revenue analysis was performed to identify periods of high customer activity.

This information can support campaign scheduling and operational planning.

6. RFM Customer Analysis

6.1 Introduction to RFM

RFM analysis is a customer analytics technique based on three dimensions:

- Recency
- Frequency
- Monetary

6.2 Recency

Recency measures how recently a customer made a purchase.

A lower Recency value indicates a more recent customer interaction.

6.3 Frequency

Frequency measures how often a customer made purchases.

Higher Frequency values generally indicate stronger customer engagement.

6.4 Monetary

Monetary value represents the total amount spent by a customer.

Higher Monetary values indicate greater customer value.

6.5 RFM Calculation

For each customer, the following values were calculated:

$$Recency = SnapshotDate - LastPurchaseDate \tag{6.1}$$

$$Frequency = Number\ of\ Unique\ Orders \tag{6.2}$$

$$Monetary = \sum Revenue \tag{6.3}$$

6.6 RFM Scoring

Customers were assigned scores from 1 to 5 based on their relative position in the Recency, Frequency, and Monetary distributions.

Recency was scored in reverse order because lower recency indicates more recent activity.

7. K-Means Customer Segmentation

7.1 Introduction

K-Means clustering was used to group customers according to similar RFM behavior.

The purpose of clustering was to identify customer groups that could be targeted with different business strategies.

7.2 Data Preparation

The RFM variables were transformed and standardized before applying K-Means.

Log transformation was used to reduce the effect of highly skewed customer spending values.

The transformed features were then standardized using `StandardScaler`.

7.3 Selecting the Number of Clusters

The Elbow Method was used to evaluate different numbers of clusters.

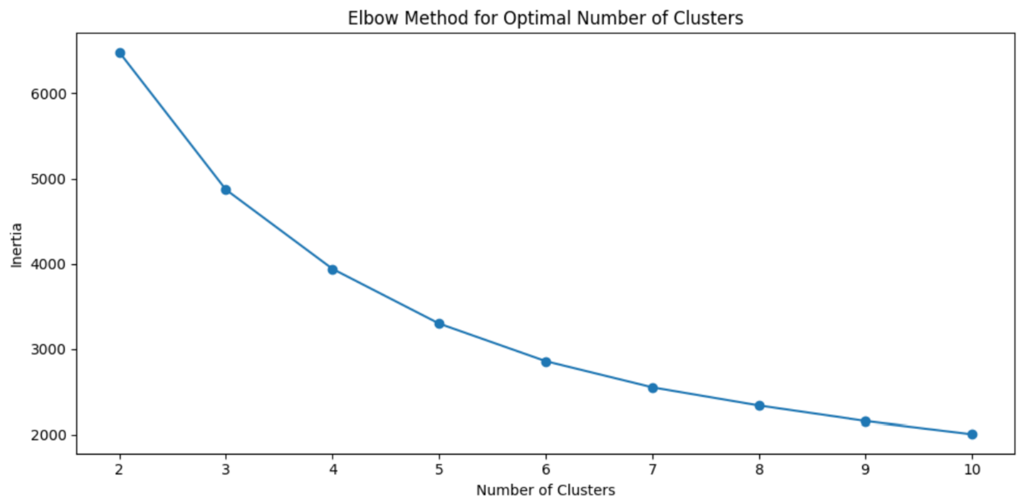

Figure 7.1: Elbow Method for K-Means

The selected number of clusters was based on the point where the reduction in within-cluster variation began to slow down.

7.4 K-Means Algorithm

The K-Means algorithm assigns each customer to the nearest cluster center and iteratively updates cluster centers.

The clustering process was performed using Scikit-learn.

7.5 Cluster Validation

The Silhouette Score was used to evaluate the quality of the clustering.

The score measures how similar customers are to their own cluster compared with other clusters.

8. Customer Segmentation Results

8.1 Segment Overview

The final customer segments were interpreted using their average Recency, Frequency, and Monetary values.

Table 8.1: Customer Segment Profile

Segment	Customers	Avg. Recency	Avg. Frequency	Avg. Monetary
Champions	–	12.17	13.75	8088.02
New Customers	–	17.70	2.19	557.32
Potential Loyalists	–	71.64	4.08	1801.78
Lost Customers	–	181.51	1.32	341.00

Note: The customer counts should be replaced with the exact values from the final `final_segment_profile` output.

8.2 Champions

Champions represent the highest-value customers.

Their average Recency is approximately 12.17 days, their average Frequency is approximately 13.75 orders, and their average Monetary value is approximately 8088.02.

These customers should receive strong retention and loyalty strategies.

8.3 New Customers

The New Customers segment contains customers with recent activity but relatively low purchase frequency and spending.

The primary goal should be to encourage these customers to make repeat purchases.

8.4 Potential Loyalists

Potential Loyalists demonstrate meaningful purchasing behavior but have not reached the highest customer-value segment.

Targeted offers, cross-selling, and loyalty incentives can help move these customers toward the Champions segment.

8.5 Lost Customers

Lost Customers have high Recency values and relatively low purchase frequency and monetary value.

These customers represent an opportunity for targeted re-engagement and win-back campaigns.

9. Dashboard Development

9.1 Dashboard Overview

A business dashboard was developed to summarize the major findings from the project.

The dashboard combines sales KPIs, product performance, geographic performance, and customer segmentation.

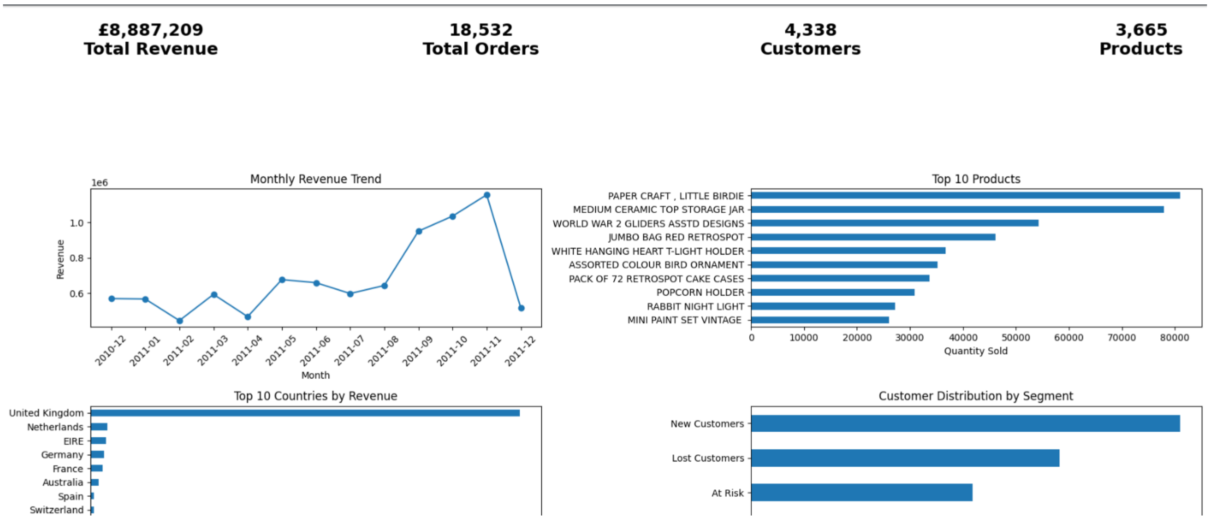

Figure 9.1: Final E-Commerce Customer Segmentation Dashboard

9.2 Key Performance Indicators

The final analysis produced the following major business metrics:

Table 9.1: Key Business Performance Indicators

Metric	**Value**
Total Revenue	£8,887,209
Total Orders	18,532
Total Customers	4,338
Total Products	3,665

The dashboard provides a consolidated view of the business and customer analytics results.

10. Key Business Insights

10.1 High-Value Customers

Champions are the most valuable customer segment, with high purchase frequency, recent activity, and high monetary contribution.

Recommendation: Develop loyalty programs, personalized offers, early-access campaigns, and premium customer service.

10.2 Customer Re-engagement

The Lost Customers segment contains customers who have not purchased recently.

Recommendation: Use targeted win-back campaigns, personalized discounts, product recommendations, and reminder campaigns.

10.3 Customer Conversion

New Customers have recent purchasing activity but relatively low frequency.

Recommendation: Encourage second purchases through follow-up offers, onboarding campaigns, and personalized product recommendations.

10.4 Potential Loyalists

Potential Loyalists have stronger customer value than low-engagement customers.

Recommendation: Use loyalty rewards, bundles, and cross-selling strategies to increase their purchasing frequency.

10.5 Geographic Concentration

The United Kingdom is the strongest revenue-generating market.

Recommendation: Continue supporting the UK market while identifying opportunities for expansion into other promising markets.

10.6 Seasonality

Revenue varies significantly across months, with stronger sales toward the later part of the analyzed period.

Recommendation: Prepare inventory and marketing campaigns before high-demand periods.

10.7 Weekly Sales Pattern

Thursday generated the highest revenue among the analyzed weekdays.

Recommendation: Consider scheduling important promotions and marketing activities around high-performing days.

11. Business Recommendations

Based on the analysis, the following recommendations are proposed.

1. **Strengthen Customer Loyalty**

   Champions should be retained using loyalty programs, personalized recommendations, exclusive offers, and premium support.

2. **Launch Win-Back Campaigns**

   Inactive customers should receive targeted re-engagement campaigns designed to encourage them to return.

3. **Increase Second Purchases**

   New customers should receive targeted offers that encourage their second purchase.

4. **Promote Cross-Selling**

   Customers with moderate purchase activity can be encouraged to buy complementary products.

5. **Use Data-Driven Campaign Timing**

   Marketing campaigns should be aligned with high-performing weekdays and seasonal demand periods.

6. **Optimize Inventory Planning**

   Product demand and seasonal patterns should be used to improve inventory planning and reduce stock-out risks.

7. **Expand Internationally**

   The company can explore opportunities in countries outside the dominant UK market to reduce geographic concentration.

12. Conclusion

This project demonstrated a complete data analytics workflow for e-commerce transaction data.

The project began with data cleaning and preprocessing, followed by feature engineering and exploratory data analysis.

RFM analysis was then used to understand customer purchasing behavior, while K-Means clustering was applied to identify groups of customers with similar characteristics.

The final customer segments provided useful insights into customer value and engagement.

The dashboard consolidated the main business metrics and analytical findings into a single visual report.

Overall, the project demonstrates how data analytics and machine learning can support customer retention, targeted marketing, inventory planning, and revenue optimization.

13. Future Scope

The project can be extended in several ways.

- Develop an interactive Power BI or Tableau dashboard.
- Implement automated customer segmentation pipelines.
- Apply customer lifetime value prediction.
- Develop product recommendation systems.
- Use time-series forecasting for future sales.
- Apply advanced clustering algorithms for comparison.
- Develop automated marketing campaign recommendations.
- Integrate real-time transaction data.

References

1. Python Software Foundation. Python Documentation.
2. Pandas Documentation. Data Analysis and Manipulation Library.
3. Scikit-learn Documentation. Machine Learning in Python.
4. Matplotlib Documentation. Visualization Library.
5. RFM Analysis methodology for customer segmentation.
6. Online Retail Transaction Dataset used for academic and analytical purposes.

A. Important Python Libraries

The major Python libraries used in the project were:

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

B. Project Deliverables

The final project deliverables include:

- Jupyter Notebook
- Customer segmentation CSV file
- Final dashboard image
- Internship project report
- Project documentation